Supplementary Material to

Measuring Economic Risk in Africa: The African Economic Risk Index and African Risk Score

October 2026

Abstract

This supplement provides material that supports the measurement and empirical results reported in the paper without interrupting its main argument. It records the domain definitions used for the eight risk pillars, the PCA conversion used for the bounded composite measures, detailed PVAR estimation and inference, expanded exact-text deduplication results, additional external-identification results, and country-level AERI and ARS outputs. The supplement uses the same analytical vintage and definitions as the paper. Sections S8–S13 contain material moved from the paper to meet the journal's length limit: an extended literature discussion, the keyword taxonomy with a worked scoring example, the pillar PCA weights, additional descriptive exhibits, the data-collection architecture, the archive-depth diagnostic and the external-identification diagnostics.

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S1 Detailed Pillar Descriptions

The main paper condenses the eight pillar descriptions. The definitions below record the domain interpretation used in the analytical vintage. Pillars are non-exclusive: one article can contribute to several domains when its economic content spans them.

S1.1 Inflation Risk

Inflation-related reporting covers price pressures and their implications for households and firms. Its mean pillar index is 4.34 points across eligible country-months. Sustained increases can reflect greater attention to input costs, purchasing power, administered prices or anticipated price adjustment. The positive and persistent response of headline inflation to an inflation-risk innovation in the paper provides an external check on this interpretation.

S1.2 FX and External Risk

External-risk reporting covers exchange-rate conditions, foreign-currency availability and external financing. Its mean pillar index is 5.03 points. Currency conditions affect imported-input costs and access to external finance, so overlap with inflation and financial-sector reporting is economically natural. The reverse-response evidence in the paper also shows that realised depreciation is followed by a higher aggregate AERI reading.

S1.3 Debt and Fiscal Risk

Fiscal reporting covers public borrowing, debt service, revenue pressure and constraints on government finances. Its mean pillar index is 8.64 points. Fiscal concerns often coincide with financial-sector pressure when public financing conditions affect domestic intermediaries and with external risk when foreign-currency obligations become more salient. The overlapping classification preserves those connections within the news record.

S1.4 Financial-Sector Risk

Financial-sector reporting covers intermediation, liquidity, balance-sheet pressure and financial stability. Its mean pillar index is 11.46 points, the largest pooled pillar mean in this vintage. The level describes the prominence of financial-sector concerns in the retained corpus and is interpreted together with within-pillar changes over time.

S1.5 Growth and Real-Economy Risk

Real-economy reporting covers production, employment, demand and broader economic activity. Its mean pillar index is 3.17 points. News can register weakening demand or production constraints before a complete statistical account becomes available and can also respond to realised changes in

activity. The reverse-response analysis in the paper shows how the aggregate AERI responds to a monthly GDP-growth innovation.

S1.6 Commodity and Supply Risk

Commodity and supply reporting covers resource production, logistics and disruptions to supply. Its mean pillar index is 0.09 points. The low pooled mean reflects the concentration of such reporting in particular countries and episodes. For interpretation, changes within exposed economies are more informative than the continental average alone.

S1.7 Political-Economy Risk

Political-economy reporting covers governance, policy implementation and their economic consequences. Its mean pillar index is 4.94 points. Political content enters the framework when it carries economic relevance and scored risk content, allowing governance and implementation concerns to be identified within broader high-risk episodes.

S1.8 Climate and Disaster Risk

Climate and disaster reporting covers environmental disruptions and their economic effects. Its mean pillar index is 1.25 points. Such events are geographically concentrated and irregular, so the pillar is most useful as a domain-specific diagnostic read alongside country context and event data.

S2 Formal PCA Conversion and Aggregation Details

Let X_j denote a bounded input, with calibration mean μ_j, standard deviation s_j and oriented first-component loading v_j. For standardized inputs $Z_j = (X_j - \mu_j)/s_j$,

$$v'Z = \sum_j \frac{v_j}{s_j} X_j - \sum_j \frac{v_j \mu_j}{s_j}. \tag{S1}$$

Define $D = \sum_j v_j/s_j > 0$ and $w_j = (v_j/s_j)/D$. Then

$$100 \sum_j w_j X_j = \frac{100}{D} \left(v'Z + \sum_j \frac{v_j \mu_j}{s_j} \right). \tag{S2}$$

The bounded-input score is therefore an affine rescaling of the first principal component. The implementation rejects calibration standard deviations below 10^{-10} and materially negative oriented loadings below -10^{-9}; negligible numerical negatives are set to zero before normalization. Calculations use the unrounded weights reported in the project outputs.

Annual AERI and raw pillar indices weight monthly values by retained article counts so that annual numerators and denominators correspond to the same article universe. Annual breadth is

the ratio of summed relevant articles to summed retained articles, while annual severity weights monthly conditional severity by relevant-article counts. When annual ARS or the eight-pillar PCA composite is required, the nonlinear composite is recomputed from annual inputs rather than formed as an average of monthly composite scores.

S3 Detailed PVAR Specification and Inference

S3.1 Macroeconomic Data and Pooled Panel Estimation

The empirical assessment relates monthly news-based risk to headline inflation, exchange-rate depreciation and monthly GDP growth. Its purpose is to test whether the measured domains display economically interpretable dynamics. ARS calibration is completed independently of these macroeconomic outcomes, preserving a clean separation between index construction and subsequent validation.

S3.1.1 Monthly variables and sample construction

Headline inflation, π_{ct}, is the supplied monthly observation of the inflation rate. The exchange rate, e_{ct}, is expressed as local-currency units per US dollar. Monthly GDP, G_{ct}, is the supplied purchasing-power-parity GDP series. Depreciation and monthly GDP growth are defined as

$$d_{ct} = 100[\log e_{ct} - \log e_{c,t-1}], \qquad g_{ct} = 100[\log G_{ct} - \log G_{c,t-1}]. \tag{S3}$$

Thus a positive d_{ct} denotes depreciation, whereas a positive g_{ct} denotes monthly GDP growth. Headline inflation retains the definition supplied in the macroeconomic panel, while depreciation and GDP growth are the log-difference transformations above. The sample-specific winsorisation bounds are reported below.

The main validation specification uses one news-based risk measure at a time—the inflation-risk pillar, the FX and external-risk pillar, AERI or ARS—together with inflation, depreciation and monthly GDP growth. The estimation sample is distinct from the descriptive sample defined by the 100-article threshold. It begins with the 36-country union selected by the prior correlation screen, then retains country-months with the risk measure, the required macroeconomic inputs and the full lag structure, subject to at least 24 usable observations per country.[1] The common GDP-augmented panel contains 4,286 observations for 36 countries and runs from May 2015 to December 2025. The 100-article rule continues to govern the principal descriptive country comparisons and ARS calibration; it is not imposed on the PVAR estimation panel. This distinction explains why a country can appear in the PVAR sample even when few or no months satisfy the descriptive coverage threshold. Using the direct pillar indices in this specification keeps the economic interpretation

[1] The estimation panel winsorises the supplied macroeconomic transformations at pooled first and ninety-ninth percentiles. In the GDP-augmented sample the retained bounds are $[-2.565, 50.934]$ for inflation, $[-5.736, 14.926]$ for depreciation and $[-0.819, 1.506]$ for monthly GDP growth.

aligned with the measurement framework in the main text. Supporting transformations and country-level estimation diagnostics are available from the authors on request.

Risk innovations are put on a comparable within-country scale. Let $\bar{r}_c$ be the country mean in the supplied scaling panel and $s_{r,w}$ the sample standard deviation of $r_{ct} - \bar{r}_c$ pooled across that panel. We use $\widetilde{r}_{ct} = r_{ct}/s_{r,w}$; country effects subsequently absorb level differences. A one-unit risk innovation therefore represents one pooled within-country standard deviation, rather than one raw index point. Scaling parameters are fixed for the reported estimates and bootstrap exercises.

S3.1.2 The pooled dynamic system

For m endogenous variables, the maintained model is

$$y_{ct} = \alpha_c + \lambda_t + \sum_{\ell=1}^{p} B_\ell y_{c,t-\ell} + \varepsilon_{ct}, \qquad E(\varepsilon_{ct}\varepsilon_{ct}') = \Sigma. \tag{S4}$$

Here α_c captures persistent country differences, and λ_t is a full year-month effect that absorbs shocks common to countries in a given month. The year-month effects therefore absorb common calendar-time variation rather than recurring month-of-year seasonality alone. The slope matrices B_ℓ are pooled across countries. In the three-variable system, $y_{ct} = (\widetilde{r}_{ct}, \pi_{ct}, d_{ct})'$. In the GDP-augmented system, $y_{ct} = (\widetilde{r}_{ct}, \pi_{ct}, d_{ct}, g_{ct})'$. In particular, the inflation equation in the latter system is

$$\begin{aligned}\pi_{ct} = \alpha_c^\pi + \lambda_t^\pi + \sum_{\ell=1}^{p}\big(&b_{\pi r,\ell}\widetilde{r}_{c,t-\ell} + b_{\pi\pi,\ell}\pi_{c,t-\ell} \\ &+ b_{\pi d,\ell}d_{c,t-\ell} + b_{\pi g,\ell}g_{c,t-\ell}\big) + \varepsilon_{ct}^\pi.\end{aligned} \tag{S5}$$

The risk, depreciation and GDP equations have the same lag structure with equation-specific coefficients. Each outcome can therefore respond to its own history and to the histories of the other variables. Pooling provides a common dynamic description conditional on fixed effects. The multivariate panel structure follows the empirical framework in Love and Zicchino (2006) and Abrigo and Love (2016), implemented here with within-transformed pooled least squares.

Let X stack all mp lagged regressors and Y stack the contemporaneous outcomes on the common estimation sample. Alternating country and year-month projections yield $\ddot{X}$ and $\ddot{Y}$. The pooled estimator and residual covariance are

$$\widehat{B} = (\ddot{X}'\ddot{X})^{+}\ddot{X}'\ddot{Y}, \quad \widehat{E} = \ddot{Y} - \ddot{X}\widehat{B}, \quad \widehat{\Sigma} = \frac{\widehat{E}'\widehat{E}}{n - mp}. \tag{S6}$$

The superscript $+$ denotes the Moore–Penrose inverse; the covariance denominator is the implemented lag-regressor adjustment. A common maximum-lag sample is used when comparing candidate orders, preventing changes in the available observations from driving lag choice. Among stable candidates

with $p \in \{1, 2, 3\}$, the criterion is

$$\mathrm{BIC}(p) = \log |\widehat{\Sigma}_p| + \frac{\log n}{n} m^2 p. \tag{S7}$$

Stability requires the maximum absolute companion-matrix eigenvalue to lie below the numerical threshold of 0.999. All reported specifications select three lags. Table S1 reports observations, country counts, roots and accepted bootstrap draws. Fixed-effects estimation with lagged outcomes can retain finite-time bias (Nickell, 1981); the long monthly dimension reduces some finite-sample concerns but leaves this source of bias relevant to interpretation.

S3.1.3 Identification and impulse responses

Let $\widehat{\Sigma} = LL'$ with L lower triangular. Forward systems order risk first, followed by inflation, depreciation and monthly GDP growth where included. This permits a risk innovation to affect the macroeconomic variables within the month, while assigning contemporaneous feedback into risk to subsequent periods. For reverse responses, the order is inflation, depreciation, monthly GDP growth and AERI. The two orderings answer complementary recursive questions about forward risk propagation and macroeconomic feedback.

The moving-average matrices and normalised responses are

$$\Phi_0 = I_m, \qquad \Phi_h = \sum_{\ell=1}^{\min(p,h)} \widehat{B}_\ell \Phi_{h-\ell}, \tag{S8}$$

$$\theta_{kj}(h) = \frac{e_k' \Phi_h L e_j}{e_j' L e_j}, \qquad h = 0, \ldots, 12, \tag{S9}$$

where e_j is the jth coordinate vector, distinct from the exchange-rate notation in equation (S3). The denominator normalises the innovation to a one-unit contemporaneous increase in the selected variable. For forward risk innovations, that unit is one within-country standard deviation. For reverse macroeconomic innovations, it is one percentage point. This convention makes the reported response magnitudes explicit while preserving the natural units used for the macroeconomic innovations.

S3.1.4 Inference and interpretation

Uncertainty bands are constructed by resampling countries with replacement, retaining each selected country's complete time history, and re-estimating the fixed-effects system. The lag order, measurement transformations and fixed ARS calibration remain unchanged across bootstrap replications. Four hundred accepted stable replications provide the fifth and ninety-fifth percentiles at each horizon. These are pointwise 90 percent intervals, so each band describes uncertainty at a given horizon. Simultaneous inference for the full response path or for the location of its population peak would require a different procedure.

Joint lag tests complement the IRFs. For example, the inflation equation tests

$$H_0 : b_{\pi r,1} = \cdots = b_{\pi r,p} = 0, \qquad W = (R\widehat{\beta})'[R\widehat{V}R']^{+}(R\widehat{\beta}), \tag{S10}$$

using country-cluster covariance $\widehat{V}$ and a χ^2_p reference distribution. The exact covariance calculation is given in Appendix S3. Rejection indicates incremental information in the included lagged variable conditional on the other lags and fixed effects. The recursive response additionally reflects contemporaneous residual covariance and propagation through the full system, so the tables report both forms of evidence.

Table S1: Pooled PVAR samples and stability diagnostics

System	Variables	Countries	Obs.	Lags	Max. root	Draws
Inflation risk + GDP	4	36	4,286	3	0.9228	400/400
FX/external risk + GDP	4	36	4,286	3	0.9222	400/400
AERI + GDP	4	36	4,286	3	0.9198	400/400
ARS + GDP	4	36	4,286	3	0.9190	400/400

Notes: All samples span May 2015–December 2025 and use the same 36-country correlation-selected panel. The PVAR sample is formed from complete model inputs and lags with at least 24 usable observations per country; the 100-article rule used for descriptive country comparisons is not imposed here. Draws report accepted/attempted country-cluster replications. A companion root below one indicates fitted dynamic stability; identification is assessed separately through the stated recursive and external-shock designs.

The 36-country set is the inclusive union of countries whose earlier country-level screening exercise produced an absolute correlation of at least 0.30 for either inflation-risk versus inflation or FX/external-risk versus the exchange rate. Twenty-eight countries enter through the inflation screen, fourteen through the exchange-rate screen and six through both. The set comprises Algeria, Angola, Botswana, Burkina Faso, Burundi, Cameroon, Cape Verde, Comoros, Democratic Republic of the Congo, Egypt, Ethiopia, Gambia, Ghana, Guinea, Guinea-Bissau, Ivory Coast, Lesotho, Liberia, Libya, Malawi, Mali, Mauritius, Morocco, Mozambique, Namibia, Nigeria, Sao Tome and Principe, Senegal, Sierra Leone, Somalia, South Africa, Tanzania, Tunisia, Uganda, Zambia and Zimbabwe.

The PVAR estimation sample should be read separately from the descriptive coverage sample reported later in this supplement. AERI and the pillar series are available for lower-volume country-months, whereas the 100-article threshold is used to define the principal descriptive comparisons and the ARS calibration sample. The PVAR retains the correlation-selected countries with complete model inputs and the required lag structure and imposes a minimum of 24 usable observations per country. This is why Guinea-Bissau and Sao Tome and Principe can enter the PVAR even though Table S7 reports zero and twenty months, respectively, under the stricter 100-article descriptive rule.

For equation k, write the country-cluster score as $s_{ck} = \ddot{X}'_c\widehat{\varepsilon}_{ck}$, let $q = mp$, and denote the

number of countries by C. The covariance used in the joint lag tests is

$$\widehat{V}_k = \frac{C}{C-1}\frac{n-1}{n-q}(\ddot{X}'\ddot{X})^{+}\left(\sum_{c=1}^{C} s_{ck}s_{ck}'\right)(\ddot{X}'\ddot{X})^{+}. \tag{S11}$$

Country resampling preserves dependence within each country's observed history. Year-month effects absorb common calendar-time variation. Inference is conditional on the correlation-screened country set and on the fixed ARS calibration.

The 36-country GDP-augmented system contains 4,286 common-sample observations, and its stability roots lie between 0.919 and 0.923. Selection is part of the design, and inference is conditional on it.

S4 Exact-Text Deduplication: Expanded Results

Repeated publication can alter a country-month index when identical high-scoring text appears in several records. The robustness exercise decodes article bodies, restricts comparison to bodies of at least 200 characters, and collapses exact matches within each country-month. Appearances in different countries or different months remain separate observations. Edited stories, translations and rewritten versions also remain separate because the test is deliberately exact.

The reconstruction covers 21,303,020 frozen-vintage records in 7,126 country-months. It is 553 records and one Togo country-month short of the archived aggregate because 42 source files were empty or unreadable. Where a reconstructed month does not fully match the archived count, unmatched frozen-vintage records remain in the denominator and only directly observed duplicates are removed. This preserves the archived universe while allowing the contribution of verified repetitions to be measured.

Table S2: Sensitivity to within-country-month exact-text deduplication

Statistic	Result
Records removed	497,091
Removed share (percent)	2.33
Cross-outlet copies removed	186,379
AERI Pearson correlation	0.9296
AERI rank correlation	0.9883
AERI mean absolute change	2.32
AERI 95th-percentile absolute change	8.30
ARS Pearson correlation	0.9680
ARS mean absolute change	0.61
Months falling below 100 articles	69

Notes: The comparison uses 6,418 country-months with at least 100 articles before and after deduplication. Exact duplicates have the same decoded body within a country-month; bodies shorter than 200 characters are retained. Both numerator and denominator are adjusted. ARS uses the original unrounded PCA weights.

Figure S1 compares the monthly cross-country median AERI before and after exact-text removal on the common coverage sample. The two paths retain the same broad chronology and both peak in

April 2020. The differences are concentrated in particular countries and months rather than spread evenly through the panel.

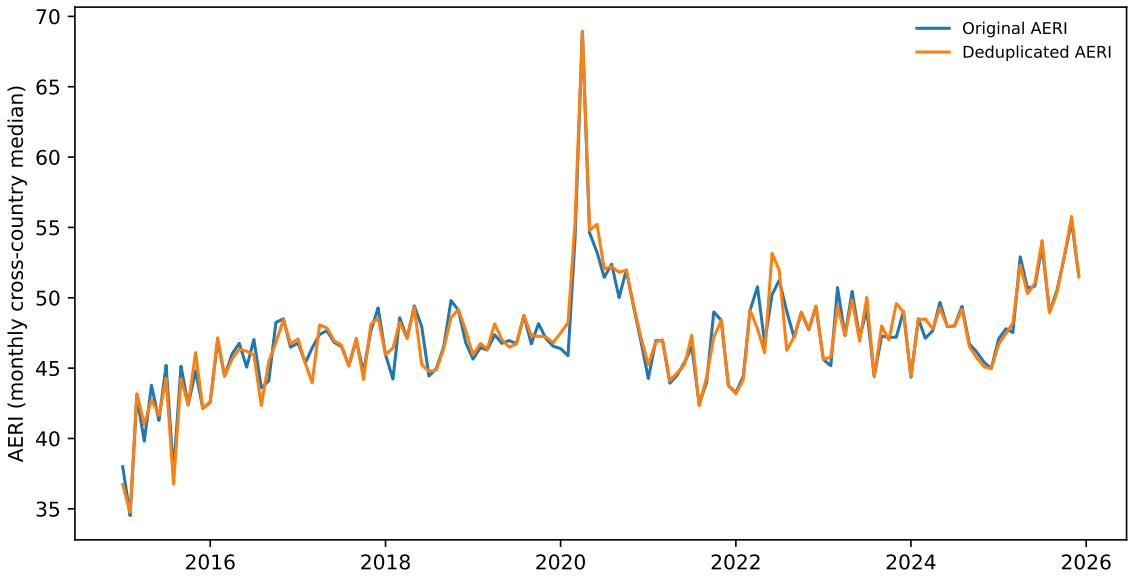


Figure S1: Monthly median AERI before and after within-country-month exact-text deduplication
Notes: Each line is the cross-country median among country-months in the common coverage sample used for the deduplication comparison. The aggregate peak remains April 2020.

Table S3: Country-level sensitivity of AERI peaks to exact-text deduplication

Country	Original	Dedup. at original	Δ	Within-country r	Dedup. peak
Sierra Leone	269.72	110.77	-158.95	0.812	2016-10
Rwanda	95.43	63.35	-32.08	0.984	2020-04
Mauritius	91.64	111.79	20.15	0.998	2020-05
Angola	85.67	72.69	-12.98	0.932	2020-04
Eswatini	75.11	87.01	11.89	0.908	2020-04
Zambia	121.49	130.62	9.12	0.684	2024-09
Cameroon	68.83	60.92	-7.91	0.992	2016-12
Botswana	90.44	85.72	-4.72	0.972	2025-12

Notes: Countries shown are those with the largest absolute change at the original peak, augmented to retain Rwanda and Mauritius because their peak behavior is discussed in the paper. The correlation is computed over the common within-country monthly sample. A change can be positive because removing duplicate records changes both the score numerator and the all-article denominator.

The median within-country AERI correlation is 0.9992 and 48 of 53 countries retain the same peak month on the common sample. Sierra Leone is the largest exception: its November 2016 observation falls from 269.72 to 110.77 and the deduplicated peak moves to October 2016. Rwanda's August 2021 observation falls from 95.43 to 63.35, while Mauritius rises from 91.64 to 111.79 in May 2020 because removal changes both the numerator and denominator. These cases identify where syndicated or repeated identical text materially affects a country history while leaving the continental chronology largely unchanged.

S5 External Monetary-Policy Exposure: Additional Results

The external-identification exercise uses the pure U.S. monetary-policy surprise from Jarociński and Karadi (2020), aggregated to the month and standardized over the 2015–2025 sample. Predetermined exposure is the 2014 stock of external debt as a share of GNI from the World Bank, transformed as $\log(1+x)$ and standardized. For country i and horizon h, the preferred local projection is

$$\begin{aligned}\text{AERI}_{i,t+h} - \text{AERI}_{i,t-1} = \alpha_i^h + \gamma_t^h + \beta_h(MP_t \times E_i^{2014}) \\ + \sum_{\ell=1}^{3} \theta_{h\ell} \Delta\text{AERI}_{i,t-\ell} + \sum_{\ell=1}^{3} \psi_{h\ell}(MP_{t-\ell} \times E_i^{2014}) + \varepsilon_{i,t+h}.\end{aligned} \tag{S12}$$

Country effects absorb persistent differences in average AERI and year-month effects absorb shocks shared by the African sample in a given month. The identifying variation is the cross-country difference in predetermined exposure to the same externally measured tightening. Standard errors are two-way clustered by country and month, and finite-cluster inference uses 28 denominator degrees of freedom for the 29-country cross-section.

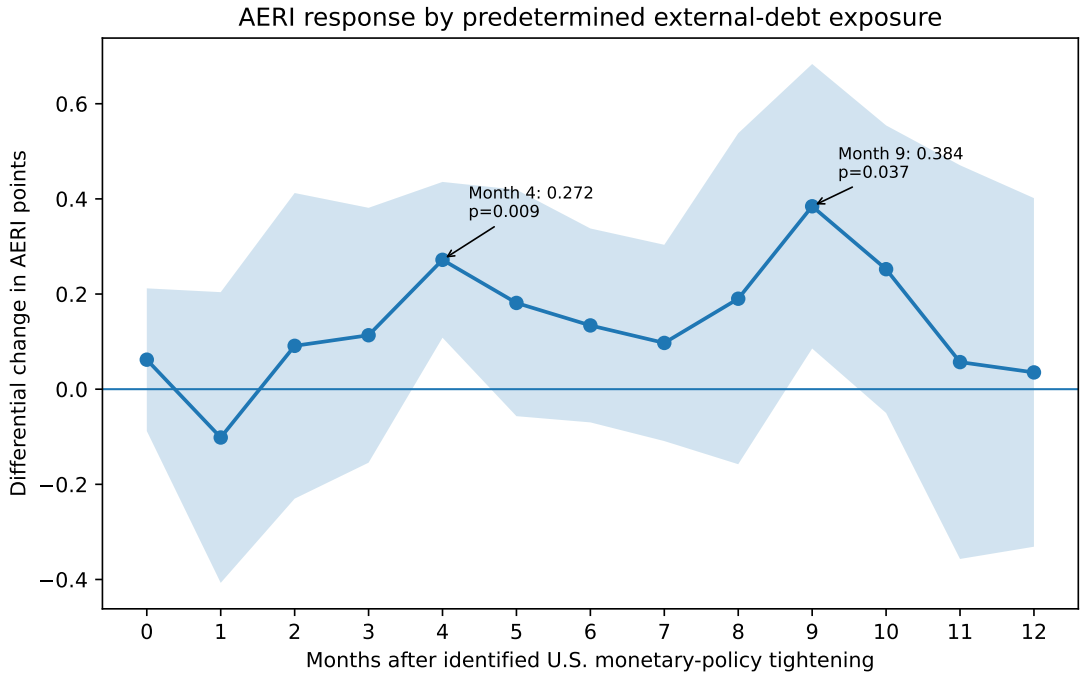


Figure S2: Differential AERI response to an externally identified U.S. monetary-policy tightening

Notes: The line reports $\widehat{\beta}_h$. Shading gives pointwise 90 percent finite-cluster intervals. Exposure is standardized $\log(1+$2014 external debt/GNI$)$.

The horizon path is delayed. The differential response reaches 0.272 AERI points at month four, with a 90 percent interval of $[0.108, 0.436]$, and 0.384 points at month nine, with interval $[0.085, 0.684]$. The six reported robustness variants retain positive month-four and month-nine coefficients. Leave-one-country-out estimates remain positive across all 29 omissions. Pre-shock

Table S4: Full horizon path for the external monetary-policy exposure specification

Horizon	$\hat{\beta}_h$	90% CI lower	90% CI upper	p-value
0	0.062	-0.088	0.212	0.487
1	-0.102	-0.407	0.204	0.576
2	0.091	-0.230	0.412	0.633
3	0.113	-0.155	0.381	0.478
4	0.272	0.108	0.436	0.009
5	0.181	-0.057	0.419	0.206
6	0.134	-0.070	0.338	0.273
7	0.097	-0.109	0.304	0.430
8	0.190	-0.158	0.538	0.360
9	0.384	0.085	0.684	0.037
10	0.252	-0.050	0.554	0.167
11	0.057	-0.357	0.471	0.817
12	0.035	-0.331	0.402	0.871

Notes: The coefficient is the differential cumulative AERI response to a one-standard-deviation U.S. monetary-policy tightening for a country one standard deviation higher in predetermined 2014 external-debt exposure. Intervals and p-values use the finite-cluster reference with 28 denominator degrees of freedom.

Table S5: Externally identified monetary-policy exposure validation

Specification	$\hat{\beta}_4$	p_4	$\hat{\beta}_9$	p_9
Primary: log debt/GNI	0.272	0.0086	0.384	0.0373
Raw debt/GNI	0.203	0.0152	0.289	0.0556
95th-percentile-capped debt/GNI	0.230	0.0284	0.347	0.0566
Exclude Mozambique	0.265	0.0305	0.390	0.0610
African-country panel restricted to FOMC announcement months	0.299	0.0078	0.354	0.0395
Alternative monetary-policy shock	0.251	0.0068	0.420	0.0534
Leave-one-country-out range	[0.214, 0.308]	29/29 $p < 0.10$	[0.282, 0.455]	29/29 $p < 0.10$

Notes: The dependent variable is the cumulative change in AERI from month $t-1$ through horizon h. The shock is the standardized pure U.S. monetary-policy surprise from the Jarociński–Karadi high-frequency decomposition, aggregated to the month. Predetermined exposure is 2014 external debt stocks as a share of GNI from the World Bank, transformed as $\log(1+x)$ and standardized in the primary specification. All regressions include country and year-month fixed effects, three lags of the AERI change and three lags of the shock–exposure interaction. Standard errors are two-way clustered by country and month. Reported p-values use a finite-cluster t reference with degrees of freedom equal to the number of African-country clusters minus one. Restricted wild-cluster-bootstrap p-values for the primary month-four and month-nine coefficients are 0.042 and 0.034, respectively. In the leave-one-country-out exercise, 28 of 29 month-four estimates and 24 of 29 month-nine estimates also satisfy $p < 0.05$.

Table S6: External-identification diagnostics and falsification checks

Check	Statistic / range	Interpretation
Pre-shock AERI-change placebos (1–6 months)	$p \in [0.156, 0.807]$	None significant at 10 percent
CBI shock × exposure, month 4	$p = 0.153$	p-value above 0.10
CBI shock × exposure, month 9	$p = 0.125$	p-value above 0.10
Joint future-shock interaction placebo	$\chi^2(6) = 11.30$, asymptotic $p = 0.079$	Large-sample reference only
Finite-cluster joint reference	$F(6, 28) = 1.884$, $p = 0.119$	p-value above 0.10
Restricted wild-cluster bootstrap	$p = 0.221$	p-value above 0.10

Notes: The joint future-shock placebo includes the contemporaneous monetary-policy-shock–exposure interaction as a control. The asymptotic Wald reference is reported for transparency, but the preferred joint inference recognizes that the identifying cross-section contains 29 African-country clusters. The finite-cluster F reference uses 28 denominator degrees of freedom; the restricted wild-cluster bootstrap uses 999 Rademacher replications by African country. The pre-shock outcome placebos are estimated without lagged dependent-variable controls that would mechanically span the placebo outcome.

outcomes and the central-bank-information component of the same announcements do not reproduce the exposure pattern, while the finite-cluster and restricted wild-cluster references place the joint future-shock placebo above conventional significance thresholds. Complete monthly shock, exposure and intermediate estimation files are available from the authors on request.

S6 All-Country AERI and ARS Outputs

Table S7 reports compact AERI and ARS outputs for all 54 economies in the 2015–2025 panel. The month count refers only to country-months with at least 100 retained articles. It is therefore a descriptive-coverage statistic rather than a PVAR eligibility count.

Table S7: Country-level AERI and ARS summary statistics, 2015–2025

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Algeria	132	65.02	83.44	2020-04	44.67	50.57	2020-04
Angola	132	53.45	85.67	2019-10	39.45	49.07	2020-05
Benin	132	43.38	71.98	2020-04	36.28	45.65	2020-04
Botswana	132	69.39	90.44	2025-12	43.67	49.59	2025-12
Burkina Faso	132	47.73	68.40	2020-04	37.90	46.83	2020-04
Burundi	132	49.20	84.80	2022-08	37.84	49.40	2022-08
Cameroon	132	48.06	68.83	2018-11	38.77	46.58	2018-11
Cape Verde	131	39.73	67.21	2022-07	31.79	42.38	2022-06
Central African Republic	129	62.25	116.38	2017-04	39.97	53.18	2017-04
Chad	96	55.64	103.13	2020-04	38.72	54.16	2020-04
Comoros	130	42.83	73.75	2022-06	36.58	47.39	2025-02
Republic of the Congo	132	61.63	87.31	2020-05	41.42	48.92	2020-05
Democratic Republic of the Congo	132	55.00	85.25	2020-08	38.92	48.42	2022-04
Djibouti	4	64.96	73.62	2024-05	41.17	45.06	2024-05
Egypt	132	40.55	97.11	2025-12	35.94	51.26	2025-12
Equatorial Guinea	73	21.10	45.64	2020-05	18.78	28.19	2020-05
Eritrea	132	49.29	77.55	2024-06	37.40	46.61	2021-03
Eswatini	131	39.94	75.11	2020-04	33.48	45.43	2020-04
Ethiopia	132	59.96	127.83	2019-02	41.11	60.87	2019-02
Gabon	132	54.49	300.16	2016-05	36.82	85.37	2016-05
Gambia	132	58.36	100.99	2020-04	40.33	52.48	2020-04
Ghana	132	57.11	85.70	2025-03	40.84	49.75	2025-03
Guinea	132	41.24	91.32	2018-07	35.85	50.04	2018-07
Guinea-Bissau	0	–	–	–	–	–	–

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Table S7 – continued from previous page

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Ivory Coast	132	30.22	46.87	2025-04	31.23	37.87	2020-04
Kenya	132	38.85	71.03	2025-12	35.01	44.12	2025-12
Lesotho	132	74.17	111.28	2020-04	44.79	54.89	2025-06
Liberia	118	90.79	128.07	2020-04	49.41	58.71	2016-08
Libya	132	12.77	32.94	2024-09	27.35	37.15	2024-09
Madagascar	132	23.83	53.24	2020-06	30.10	41.60	2020-06
Malawi	132	56.18	93.32	2025-02	40.11	51.12	2025-02
Mali	132	61.90	82.69	2016-09	43.35	49.62	2016-09
Mauritania	132	1.00	30.51	2019-06	2.26	31.32	2019-06
Mauritius	132	19.05	91.64	2020-05	29.70	52.02	2020-05
Morocco	132	22.67	37.66	2025-10	31.17	36.68	2020-04
Mozambique	132	34.16	61.17	2020-04	33.05	41.84	2020-04
Namibia	132	49.96	73.27	2025-12	37.97	44.78	2025-12
Niger	132	28.72	51.58	2025-12	29.69	38.50	2020-04
Nigeria	132	73.04	98.43	2024-02	46.21	52.76	2024-02
Rwanda	110	39.31	95.43	2021-08	32.79	51.79	2021-08
Sao Tome and Principe	20	33.90	44.72	2020-07	24.94	32.31	2021-12
Senegal	132	25.47	49.80	2025-02	32.15	40.24	2025-02
Seychelles	2	10.98	11.43	2025-06	20.94	21.07	2025-06
Sierra Leone	132	131.61	269.72	2016-11	58.50	79.56	2016-11
Somalia	132	8.26	48.54	2025-11	24.59	40.23	2025-11
South Africa	132	42.58	79.22	2020-04	36.71	48.74	2020-04
South Sudan	132	48.74	98.09	2025-07	37.21	51.08	2025-07
Sudan	132	5.05	17.15	2024-07	28.03	36.93	2019-02
Tanzania	132	14.88	54.45	2025-12	20.39	39.41	2025-12

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Table S7 – continued from previous page

Country	Months	Mean AERI	Max AERI	AERI peak	Mean ARS	Max ARS	ARS peak
Togo	132	72.05	108.08	2020-04	43.70	54.77	2020-04
Tunisia	132	47.51	74.22	2022-03	38.36	47.17	2022-03
Uganda	132	60.50	84.86	2025-07	41.33	47.93	2025-07
Zambia	132	96.43	121.49	2024-09	50.71	57.41	2025-06
Zimbabwe	132	64.99	98.27	2019-01	42.55	51.11	2019-01

Notes: Months count country-months with at least 100 retained articles. AERI is the headline article-score index. ARS is the fixed PCA-weighted 0–100 African Risk Score used in the main analysis. Peak months are identified within January 2015–December 2025. Guinea-Bissau has no month meeting the 100-article descriptive threshold in this vintage and is therefore shown with missing summary statistics rather than a misleading low-risk value. These month counts describe the 100-article descriptive sample and do not determine inclusion in the PVAR, which uses a separate correlation-screened, macro-data-complete estimation sample. The complete monthly panel is available from the authors on request.

S7 Individual Country AERI Dynamics

This section reports monthly AERI for the ten economies used in the annual comparison in the paper: Nigeria, Ghana, Algeria, Morocco, Egypt, Ethiopia, South Africa, Kenya, Angola and Ivory Coast. The dashed reference line in the figures is the contemporaneous median across eligible countries.

Table S8: Country-level AERI: descriptive comparisons, 2015–2025

Country	Months	Mean	SD	Peak	Peak month
Nigeria	132	73.04	8.75	98.43	2024-02
Ghana	132	57.11	10.47	85.70	2025-03
Algeria	132	65.02	6.34	83.44	2020-04
Morocco	132	22.67	7.54	37.66	2025-10
Egypt	132	40.55	11.29	97.11	2025-12
Ethiopia	132	59.96	17.50	127.83	2019-02
South Africa	132	42.58	7.61	79.22	2020-04
Kenya	132	38.85	7.02	71.03	2025-12
Angola	132	53.45	8.60	85.67	2019-10
Ivory Coast	132	30.22	5.28	46.87	2025-04

Notes: Monthly AERI uses the article-score definition reported in the paper. Only months with at least 100 retained articles are included. Means and sample standard deviations give equal weight to eligible months; they differ from article-weighted annual AERI. Peak denotes the maximum eligible monthly observation, with the earliest month reported in a tie. These ten economies form the illustrative group used in the paper; PVAR sample selection is conducted separately.

The country series combine common shocks with marked differences in levels, volatility and timing. Nigeria remains high through much of the sample and peaks in February 2024. Ethiopia has the largest standard deviation in the ten-country group and records its maximum in February 2019. South Africa and Algeria show pronounced April 2020 peaks, while Morocco's series rises more gradually later in the sample. The common vertical scale preserves these differences rather than normalizing each country to its own history.

S8 Additional Literature Discussion

This section gives the full version of the literature discussion that the paper condenses in Sections 2.1, 2.4 and 2.5.

S8.1 Economic risk, expectations and ex ante behaviour

Economic decisions depend on prospective outcomes as well as current conditions. Bloom (2009) shows how uncertainty can delay investment and hiring, while Jurado et al. (2015) measures uncertainty through the unpredictability of economic outcomes. These contributions motivate attention to information that appears before realised activity. A news-based risk index captures a different object: the salience and severity of economic concerns in reporting.

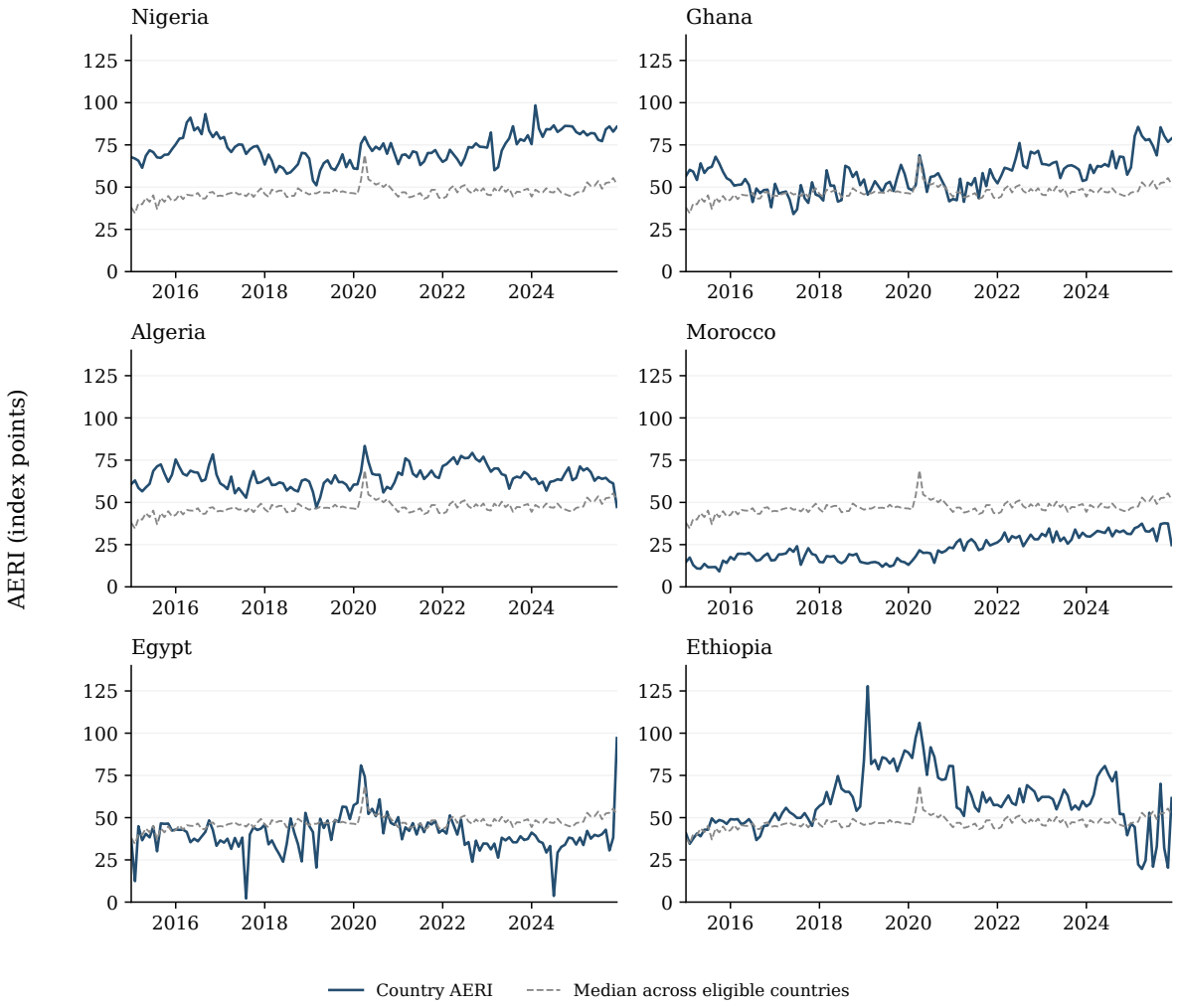


Figure S3: Monthly country AERI: Nigeria, Ghana, Algeria, Morocco, Egypt and Ethiopia
Notes: Solid lines show monthly AERI. The dashed line is the median across countries with at least 100 retained articles in the corresponding month. No smoothing or country-specific rebasing is applied.

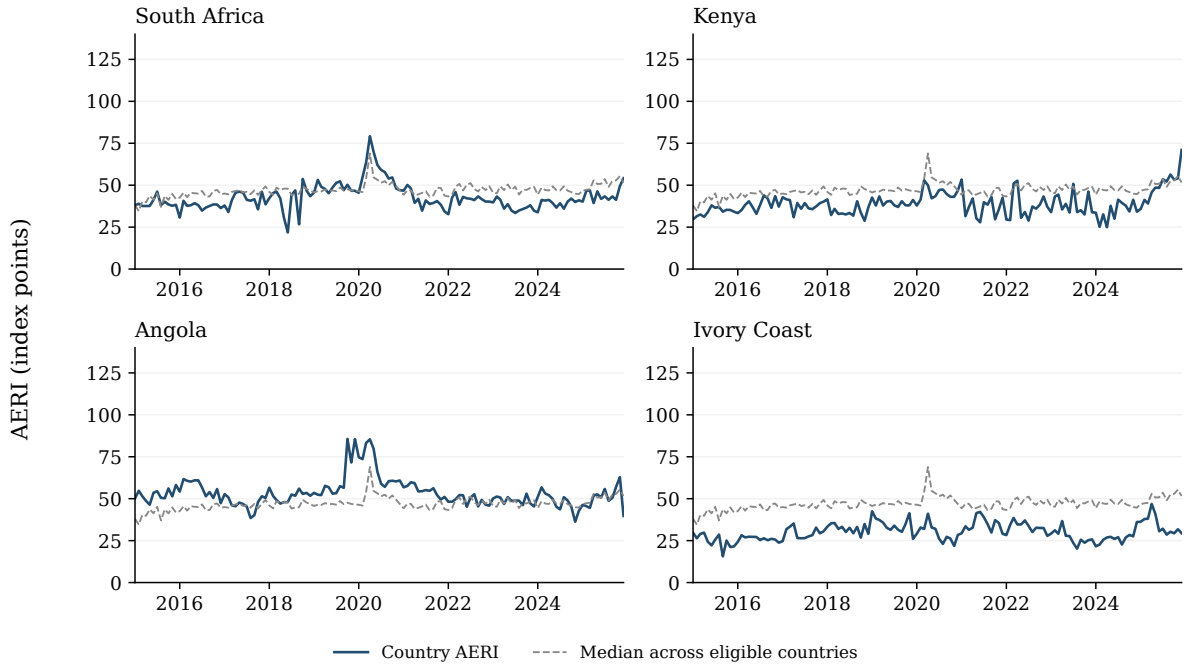


Figure S4: Monthly country AERI: South Africa, Kenya, Angola and Ivory Coast
Notes: Definitions, timeline and vertical scale match the preceding figure. The reference median varies with the composition of eligible countries.

A clearly anticipated increase in financing costs may generate substantial risk reporting even when there is little uncertainty about whether it will occur. Conversely, a highly uncertain development may attract little coverage in a thin archive. AERI is built to capture the first of these margins: the amount and intensity of economic-risk content in the observed news flow. The empirical analysis then asks whether movements in that content bear systematic relationships to realised macroeconomic conditions.

S8.2 Media, information frictions and macroeconomic beliefs

Information reaches households and firms unevenly. Mankiw and Reis (2002) show how slow information diffusion can generate persistent inflation dynamics. Shiller (2017) emphasises narratives as a mechanism through which economic interpretations spread. News records changing attention and beliefs, but it also responds to realised conditions. The empirical analysis therefore examines movements from news to macroeconomic outcomes and feedback from realised conditions to the news measures.

An inflation response after inflation-risk reporting can arise for different reasons. News may record cost pressures before their full price effects, reflect expectations about those pressures, or respond to information also used by firms. In the other direction, a rise in AERI after depreciation shows realised stress entering the news environment. The empirical design examines both relationships.

S8.3 Cross-country measurement challenges and African contexts

Cross-country indices are sensitive to publication volume, source composition, language and archive access. Scaling by total article counts removes the direct level effect of media volume, while coverage diagnostics track variation in the underlying observation base. We also distinguish missing source months from observed months with zero measured risk. These choices are especially important when one statistical procedure is applied across heterogeneous media environments.

Comparability has both a time-series and a cross-sectional dimension. Within a country, changes in the available outlet set can affect the observed news mix; across countries, similar scores can arise from media systems with different degrees of economic-news specialisation. The common scoring rule provides definitional consistency and the minimum-coverage restriction limits the influence of very small denominators. The secondary empirical exercises further absorb persistent country differences and shared calendar shocks. Time-varying changes in national source coverage remain a separate measurement concern and are examined through the coverage diagnostics.

S8.4 Aggregation design and missing observations

Our aggregation step sits within this tradition but makes different choices for a different purpose. Gentzkow et al. (2019) treat textual measurement as an empirical design problem in which preprocessing, feature definition and interpretation affect the resulting economic variable. Dictionary-based scoring allows the contribution of an article to be traced to explicit rules. PCA then reduces the dimensionality of the aggregate inputs (Jolliffe & Cadima, 2016). It determines weights from covariance, rather than choosing them to maximise correlation with inflation or output, so the composite is not fitted to the outcomes it is later compared with. Economic interpretation continues to depend on the definitions of the inputs.

PCA enters only at the aggregation stage. Economic relevance is defined beforehand through the article-scoring rules. Standardisation prevents the numerical scale of an input from determining its principal-component loading, and converting the loadings back to the bounded input scale explains why a component with a smaller standardised loading can receive a larger final weight. We report the loadings, scaling bounds and final weights so that the empirical weighting step can be reproduced directly. Missing source months are kept as missing rather than filled in by a model. A factor model in the spirit of Bańbura and Modugno (2014) could estimate those months, but the published index would then depend on the model used to estimate them. Keeping the gaps visible leaves that choice to the user and keeps a missing archive month distinct from an observed month with no measured risk.

S8.5 Alternative sources of high-frequency information

A growing literature uses alternative data to supplement conventional economic statistics where coverage is sparse or publication lags are long. Yeh et al. (2020) use publicly available satellite imagery and deep learning to measure local economic well-being across Africa. Buell et al. (2021)

combine high-frequency indicators, including search and payments data, with statistical models to track economic activity in Sub-Saharan Africa. More recently, Fotopoulou et al. (2026) examine machine-learning and satellite-data methods for nowcasting real GDP in data-constrained settings. These studies illustrate a broader measurement problem: economically relevant information often arrives at frequencies and in forms that are not captured by standard national accounts releases.

S9 Measurement Details Moved from the Paper

Table S9 gives representative terms from the production dictionaries and a worked example of how the matching rules map one article into AERI, pillar and ARS inputs. Table S10 reports the loadings, scaling bounds and weights for the eight-pillar PCA composite.

Table S9: Illustrative keyword taxonomy used in article scoring

Scoring layer	Illustrative terms from the production dictionary	Role in the computation
Economic relevance	economy, GDP, inflation, budget, debt, central bank, currency, bank, unemployment; French examples include *économie*, *PIB*, *dette*, *taux de change*; Portuguese examples include *economia*, *PIB*, *dívida*, *câmbio*	At least one match sets $R_i = 1$. Non-matching articles receive a zero article score but remain in the country-month denominator.
Risk pillars	Inflation: inflation, cost of living, shortage; FX/external: exchange rate, currency, devaluation, reserves; debt/fiscal: debt, default, restructuring, budget deficit; growth: recession, slowdown, unemployment; analogous dictionaries define the other four pillars	Pillar matches identify the domain composition of a relevant article. Pillars are non-exclusive and repeated mentions do not increase the pillar weight.
Severity tiers	Tier 1: concern, risk, uncertainty; Tier 2: surge, spike, worsening, deterioration; Tier 3: crisis, collapse, emergency; Tier 4: default, hyperinflation, bank run, economic collapse	A relevant article receives baseline severity 1; when several severity cues occur, S_i equals the highest matched tier.
Title/emphasis cues	crisis, default, devaluation, inflation, shortage, collapse, debt, restructuring, recession, bank, emergency	A title match adds 0.5 to W_i; a body of at least 1,200 normalised characters adds 0.2; two or more active pillars add 0.3. The weight is clipped to $[1, 2]$.

Notes: The table reports representative terms rather than the full dictionary. The supplied production configuration contains explicit English, French and Portuguese terms. Terms are assigned to relevance, pillar, severity and title-hint dictionaries before aggregation, independently of the subsequent macroeconomic validation outcomes.

Worked scoring example. Consider a hypothetical article titled "Inflation surge after currency devaluation." Suppose the normalised body exceeds 1,200 characters and contains no severity cue above tier 2. The article matches the economic-relevance dictionary, the inflation and FX/external

pillars, the tier-2 term "surge," and a title-risk cue. Hence $R_i = 1$, $S_i = 2$, $T_i = L_i = M_i = 1$, so

$$W_i = \text{clip}(1 + 0.5 + 0.2 + 0.3, 1, 2) = 2, \qquad A_i = R_i S_i W_i = 4.$$

If a country-month contains 100 retained articles and this is the only relevant article, this one record contributes $100(4/100) = 4$ points to AERI. The inflation and FX/external pillar indices each receive 4 points because the article contributes once to each active pillar. The same hypothetical month has breadth $B = 1/100 = 0.01$, conditional severity $V/4 = 2/4 = 0.50$, and structural spread $K/8 = 2/8 = 0.25$. With the fixed PCA ARS weights $(0.2625, 0.2722, 0.3223, 0.1430)$ and intensity $\text{AERI}/200 = 0.02$, the corresponding illustrative score is

$$\text{ARS} = 100\{0.2625(0.02) + 0.2722(0.01) + 0.3223(0.50) + 0.1430(0.25)\} = 20.49.$$

The example is purely mechanical and illustrates article-level scoring; monthly scores are formed from the full retained country-month article set.

Table S10: PCA weights, scaling bounds and loadings for the eight pillars

Pillar	Weight (%)	Log lower	Log upper	PC1 loading
Inflation	12.95	0.0000	2.9629	0.4052
FX and external	13.21	0.0000	4.2958	0.3584
Debt and fiscal	17.29	0.0000	4.3550	0.4071
Financial sector	17.30	0.0000	4.5177	0.4219
Growth and real economy	11.90	0.0000	2.6486	0.3861
Commodity and supply	6.48	0.0000	1.0248	0.1289
Political economy	9.92	0.0000	2.9873	0.3063
Climate and disaster	10.95	0.0000	2.3585	0.3227

Notes: Bounds are the first and ninety-ninth calibration percentiles of $\log(1 + P_p)$. PC1 explains 52.50 percent of standardised variation. The weights quantify each transformed pillar's contribution to the common cross-domain factor after scale adjustment; economic interpretation remains anchored in the domain-specific series and results. Calculations use unrounded weights, which sum to one.

Calibration pools eligible country-month rows without imposing equal total country weights. Countries with more eligible months consequently contribute more observations. Holding the calibration fixed makes subsequent changes in a score attributable to changed inputs rather than to month-by-month re-estimation of its weights. A future recalibration should be treated as a new measurement vintage, with the associated parameters retained. Because ARS compresses several diagnostics into one score, the individual pillar and component series remain necessary for interpretation; information outside the common movement of the four ARS inputs may still be economically important.

Figure S5 makes two comparisons. Panel A compares AERI with the eight-pillar PCA composite, which summarises common movement across the transformed domain indices. Their monthly-median correlation is 0.558 over 2015–2025, and both reach their maximum in April 2020. The common peak indicates that the dominant AERI episode coincided with unusually broad movement across

risk domains. Outside that episode the series diverge in informative ways. May 2022, for example, is the second-highest monthly median of the pillar composite but falls outside the five highest AERI months, pointing to stronger cross-domain co-movement than aggregate article-score intensity alone would suggest.

Panel B isolates the weighting choice. The benchmark 40/25/20/15 score and the PCA-weighted score use the same inputs and observations; their monthly cross-country medians correlate 0.975 and both attain their 2015–2025 maximum in April 2020. The broad ARS chronology changes little under this substantially different weighting scheme.

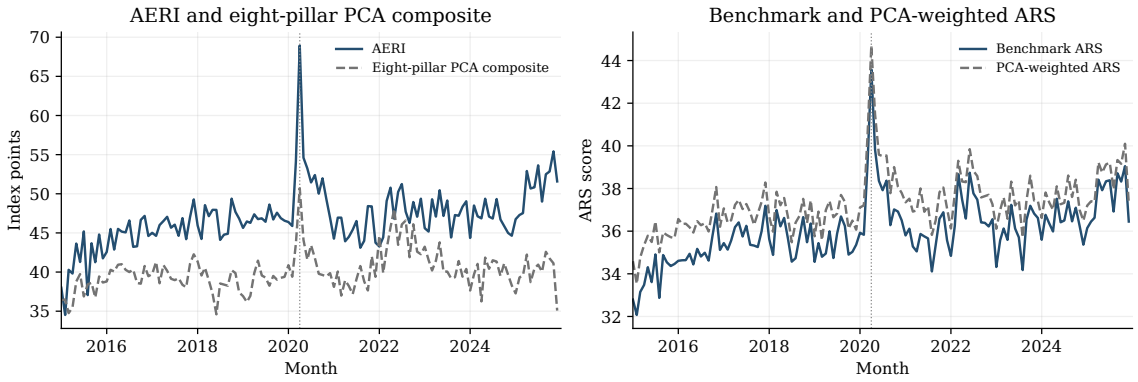


Figure S5: Weighting sensitivity of the AERI-related and ARS measures

Notes: Panel A compares AERI with the eight-pillar PCA common-factor composite. Panel B compares benchmark ARS, using 40/25/20/15 weights on intensity, breadth, severity and structure, with PCA-weighted ARS. Lines are monthly unweighted country medians for country-months with at least 100 retained articles. The vertical reference marks April 2020. The monthly-median correlations are 0.558 for AERI versus the eight-pillar PCA composite and 0.975 for benchmark versus PCA-weighted ARS.

Table 3 of the paper provides the corresponding distributional comparison. We hold the sample and component definitions fixed, so the exercise isolates the effect of the weights themselves. The PCA specification supplies a reproducible empirical weighting rule, while the 40/25/20/15 benchmark provides a simple reference point. The two schemes diverge most when intensity and severity move differently, because those components receive the largest differences in weight. A country-month with high conditional severity but only moderate AERI intensity will tend to score higher under the PCA weights; an episode dominated by unusually high intensity with less exceptional severity will tend to score higher under the benchmark.

S10 Additional Descriptive Results

Figure S6 follows annual AERI for the ten economies examined in the country discussion. The common colour scale makes differences in intensity and persistence visible without imposing a common economic explanation on each episode. Annual values aggregate the monthly article-score numerator and denominator.

Figure S6 shows that the aggregate African pattern hides marked country differences. Angola, Egypt, Ethiopia, Kenya and South Africa all record higher annual AERI in 2020 than in their

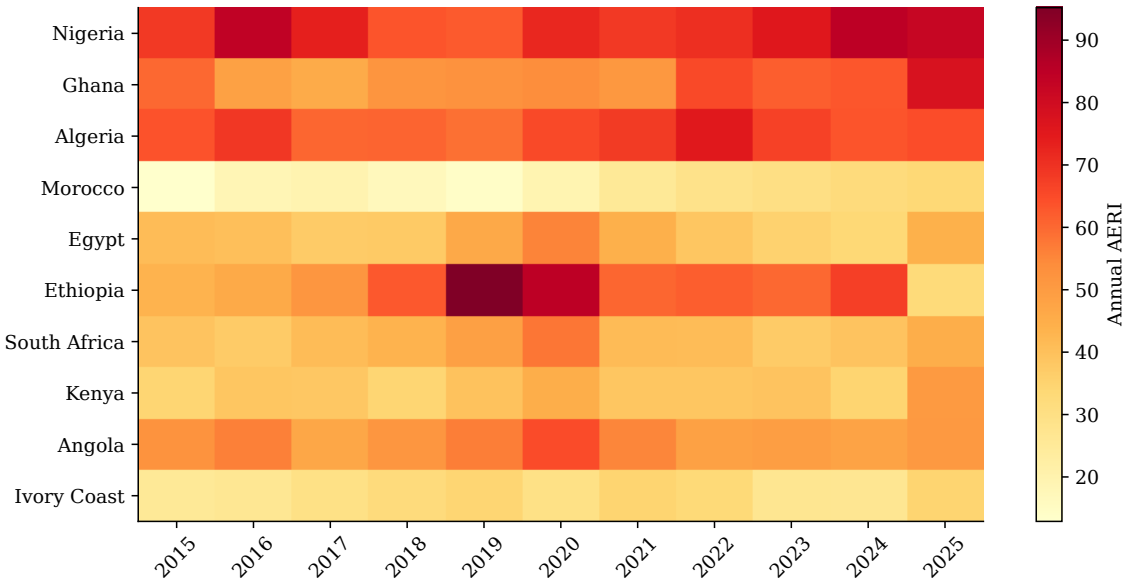


Figure S6: Annual AERI in selected African economies

Notes: The figure reports annual AERI for the ten illustrative economies over 2015–2025. Monthly AERI values are aggregated using monthly retained-article counts, so the annual statistic preserves the article-score numerator and denominator. The ten countries form the illustrative country group used in the descriptive discussion, and the underlying annual file covers all 54 economies. The common colour scale highlights substantial cross-country differences in levels, persistence and the timing of peaks.

own pre-pandemic averages. Nigeria stays comparatively high through much of the sample and reaches its annual maximum in 2024. Morocco rises gradually after 2020 rather than showing a single pandemic-centred jump, while Ethiopia peaks in 2019 and remains elevated in 2020. The continental median captures the common timing of some episodes, but it averages over persistence and country-specific peaks.

Figure S7 reports the pooled mean of each raw pillar index across the 6,488 eligible country-months, and Table S11 reports the means by subperiod.

The subperiod means in Table S11 show the same pattern. Financial-sector and debt/fiscal risk remain the two largest domains in each reported period, but the composition changes around major episodes. Growth and real-economy risk rises from 3.14 in 2015–2019 to 4.85 in 2020, while inflation risk rises from 3.47 before the pandemic to 5.33 in 2021–2025. FX/external and debt/fiscal risk are somewhat lower on average in 2021–2025 than before 2020. The pillar system distinguishes changes in the composition of risk that the aggregate AERI series alone would obscure.

S11 Data Collection Architecture

The collection and measurement sequence is: country-site registry; monthly retrieval; text normalisation; relevance classification; severity and textual emphasis weighting; country-month aggregation; pillar diagnostics; and ARS construction. A failed source month is treated as a gap in the observation system, while a retained month with no matched economic content is recorded as a measured zero.

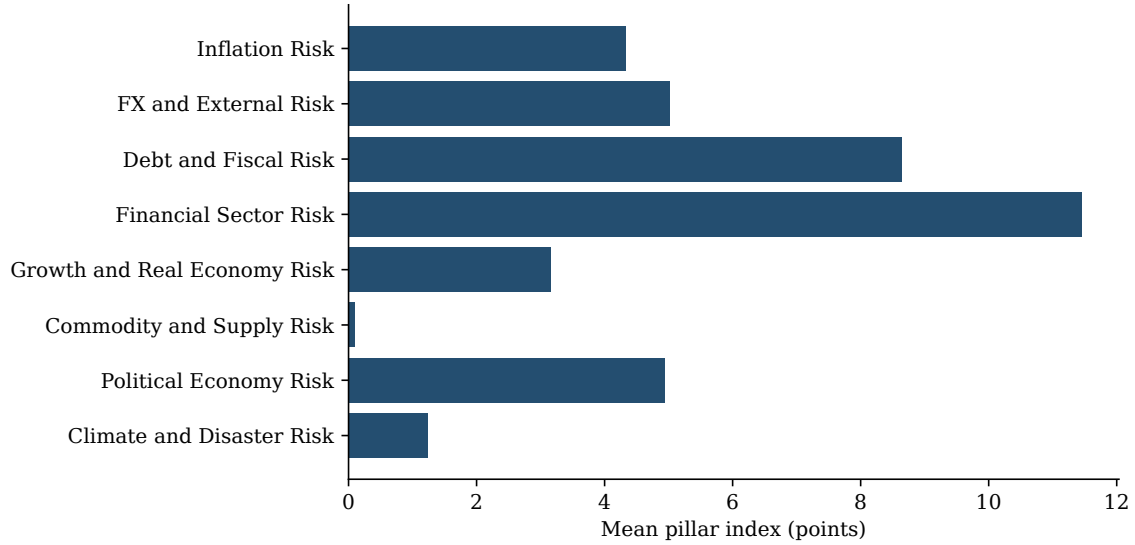


Figure S7: Average domain-specific pillar indices

Notes: Bars are ordered by magnitude and report pooled means of the eight raw pillar indices across 6,488 country-months with at least 100 retained articles. Each pillar uses the same graded article score as AERI, with the numerator restricted to articles assigned to that domain. Pillars are non-exclusive, so a single article may contribute to several domain indices. Financial-sector and debt/fiscal reporting have the largest pooled means in this vintage.

Table S11: Pillar-specific economic-risk content over time

Pillar	2015–2019	2020	2021–2025	Overall
Inflation risk	3.47	3.49	5.33	4.34
FX and external risk	5.59	5.11	4.48	5.03
Debt and fiscal risk	9.47	9.64	7.65	8.64
Financial-sector risk	11.75	12.87	10.90	11.46
Growth and real economy risk	3.14	4.85	2.85	3.17
Commodity and supply risk	0.07	0.03	0.13	0.09
Political-economy risk	5.27	5.51	4.50	4.94
Climate and disaster risk	1.25	1.11	1.27	1.25

Notes: Entries are means of the raw pillar indices on the same score-per-article scale used in AERI, computed over eligible country-months. The pillars are non-exclusive: one article may contribute to more than one domain, and the rows describe overlapping domain-specific content. The period split is descriptive and highlights changes in the composition of reported economic risk.

The replication extension adds exact-text comparison within country-months, including across outlets, and retains duplicate counts, representative examples and file fingerprints. This step addresses identical article bodies; broader questions of story independence and media representativeness remain part of the measurement limitations discussed in the main text.

The index vintage spans 131 or 132 reported months per country within 2015–2025. Annual country values are available from the authors upon request. Descriptive comparisons generally require at least 100 articles per month. The PVAR sample is defined separately: it starts from the 36-country correlation-screened set, retains country-months with complete model inputs and the required lag structure, and requires at least 24 usable observations per country. The descriptive 100-article threshold is not imposed on PVAR estimation. Formal PCA derivations, detailed pillar descriptions and expanded robustness results are reported in Sections S1–S5 of this supplement.

S12 Archive-Depth Diagnostic

Table S12 examines whether AERI moves mechanically with the number of retained articles after persistent country differences are removed. The first specification includes country fixed effects; the second adds year-month fixed effects. Both use country-clustered standard errors and the 6,488 country-months that satisfy the principal coverage threshold.

Table S12: AERI and within-country news-archive depth

Specification	Coefficient	Std. error	p-value	Observations
Country fixed effects	1.141	2.978	0.702	6,488
Country and year-month fixed effects	-1.474	3.153	0.640	6,488

Notes: The dependent variable is AERI and the regressor is the natural logarithm of retained article records in the country-month. The sample contains the 6,488 country-months with at least 100 retained articles, covering 53 countries; Guinea-Bissau has no month meeting the threshold in this vintage. Standard errors are clustered by country. The within-country correlation between AERI and log article volume is 0.035. The second specification also absorbs year-month fixed effects.

S13 External-Identification Diagnostics

Table S6 in Section S5 reports falsification and small-cluster inference checks for the externally identified monetary-policy exposure design. The joint future-shock placebo includes the contemporaneous shock–exposure interaction as a control. Pre-shock outcome placebos are estimated without lagged outcome controls that would mechanically overlap the placebo dependent variable. Complete horizon paths, robustness results, leave-one-country-out estimates, finite-cluster calculations, wild-cluster-bootstrap outputs and the underlying monthly shock and exposure files are available from the authors on request and are summarised in Section S5.

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